How Does the Model Count in French? Mapping Internal Number Representations of the Vigesimal Counting System in Large Language Models

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Abstract

Large language models encode numbers internally as base-10 digit representations, but how do they handle counting systems where the surface form encodes arithmetic rather than digits? French uses a vigesimal (base-20) system for numbers 70–99: "quatre-vingt-dix-huit" (98) literally means "four-twenty-ten-eight" ($4 \times 20 + 10 + 8$). We probe MISTRAL-7B hidden representations to ask whether the model can decompose this arithmetic structure into standard base-10 digits. Using circular digit probes across four input formats—digit strings, English words, France French words, and Belgian French words (which use a simpler decimal system)—we find that French number word representations encode base-10 digits at 98.0% accuracy, comparable to English (98.5%) and Belgian French (99.0%), and significantly higher than digit strings (91.0%). The vigesimal structure causes only localized difficulty: the *soixante-dix* (70s) range drops to 88.2%, with errors revealing partial encoding of the "soixante" (60) component rather than the composed value. Our most striking finding is that number word representations peak at early layers (5–6), while digit string representations peak at the final layer (31), suggesting fundamentally different processing pathways for semantic and tokenized numerical inputs.

1 Introduction

In the number 98, every French child learns to say "quatre-vingt-dix-huit"—literally, "four-twenty-ten-eight," encoding the arithmetic expression $4 \times 20 + 10 + 8$. This vigesimal (base-20) counting system, inherited from Gaulish and Celtic linguistic influence, is one of the most notoriously counterintuitive features of French. Meanwhile, Belgian and Swiss French speakers say simply "nonante-huit" (ninety-eight), using a transparent decimal structure. This linguistic divergence creates a natural experiment: when a language model processes these two surface forms, does it arrive at the same internal numeric representation?

Recent work has revealed that large language models encode surprisingly precise base-10 digit representations in their hidden states [Levy and Geva, 2024, Kadlčík et al., 2025]. Levy and Geva [2024] showed that circular probes can recover individual digits from number token representations with over 90% accuracy, and that these representations causally influence model outputs. However, this line of work has focused almost exclusively on digit strings and simple English word forms. Whether models can decompose the complex arithmetic structure of vigesimal French number words into clean base-10 representations remains unknown.

Why does this matter? Understanding how models process numbers embedded in complex linguistic structures reveals fundamental aspects of the interface between language and mathematics in transformers. If a model can internally "translate" "quatre-vingt-dix-huit" ($4 \times 20 + 10 + 8$) into the digit representation $[9, 8]$, it demonstrates that distributional learning has captured not just word

co-occurrences but implicit arithmetic structure. Conversely, if vigesimal forms produce degraded representations, it points to fundamental limits of how linguistic structure constrains numerical reasoning.

We apply circular digit probes [Levy and Geva, 2024] to MISTRAL-7B across four input formats: digit strings, English number words, France French number words (vigesimal), and Belgian French number words (decimal). Our probes operate on hidden representations extracted at the last token position and predict each base-10 digit via circular (cosine/sine) encodings. We evaluate on 1,000 numbers (0–999) with a stratified train/test split.

Our results reveal three key findings. First, French number word representations—including vigesimal forms—encode base-10 digits at 98.0% accuracy, comparable to English words (98.5%) and significantly outperforming digit strings (91.0%, $p = 0.002$ by McNemar's test). Second, the vigesimal structure causes only localized difficulty in the *soixante-dix* (70s) range (88.2%), with errors revealing partial encoding of the "soixante" (60) component. Third, number word representations peak at early layers (5–6), while digit string representations peak at the final layer (31), suggesting fundamentally different processing pathways.

We make the following contributions:

- We conduct the first systematic study of how LLMs represent French vigesimal numbers internally, showing that base-10 digit information is recoverable at 98.0% accuracy despite the complex arithmetic surface form.
- We exploit the natural experiment between France French (vigesimal) and Belgian French (decimal) to isolate the effect of counting system structure on internal representations, finding that Belgian French achieves perfect accuracy on vigesimal-range numbers where France French drops to 88.2%.
- We discover that number word representations emerge early in the network (layers 5–6) while digit string representations peak late (layer 31), revealing distinct processing pathways for semantic versus tokenized numerical inputs.
- We provide a detailed error analysis showing that vigesimal errors reflect partial arithmetic decomposition—the model sometimes encodes "soixante" as 60 rather than the composed value 70+—offering a window into the model's internal processing of compositional number words.

The remainder of this paper is organized as follows. Section 2 reviews related work on number representations in LLMs. Section 3 describes our probing methodology and experimental setup. Section 4 presents our main results, layer-wise analysis, and error analysis. Section 5 discusses implications and limitations. Section 6 concludes.

2 Related Work

Number representations in LLMs. A growing body of work investigates how large language models encode and process numbers. Levy and Geva [2024] demonstrated that LLMs represent numbers using per-digit circular encodings in base 10, achieving over 90% accuracy with linear probes on models including MISTRAL-7B and Llama 3. They showed that causal interventions on these digit representations modify model outputs, confirming that the representations are functionally relevant. Kadlčík et al. [2025] extended these findings across multiple model families (Llama 3, Phi 4, OLMo 2) and scales (1B–72B parameters), showing that sinusoidal probes achieve near-perfect accuracy and that embedding precision explains a large fraction of arithmetic errors. Štefánik et al. [2025] showed that different LLMs learn interchangeable, universal number representations that generalize across hidden states and input contexts. We build directly on the circular probing methodology of Levy and Geva [2024], extending it from digit strings and preliminary English word-form results to a systematic cross-lingual comparison involving French vigesimal number words.

Tokenization and numerical reasoning. How numbers are tokenized strongly influences model performance on arithmetic tasks. Singh and Strouse [2024] showed that right-to-left tokenization improves arithmetic accuracy and that model errors follow systematic patterns suggesting structured computation. Nogueira et al. [2021] found that subword tokenization fails for multi-digit addition, while position-augmented formats can enable accurate computation. Zhou et al. [2024] studied how different numeral systems (base 10, 100, 1000) affect learning efficiency, finding that base 10 is consistently more data-efficient. French vigesimal number words present an extreme case of

"unusual surface form," where the tokenized representation encodes arithmetic operations rather than digit values. Our work complements these studies by examining how tokenization of complex number words affects the quality of internal numeric representations rather than output accuracy.

Numerical reasoning and arithmetic in LLMs. The broader landscape of numerical reasoning in LLMs includes work on arithmetic circuits [Hanna et al., 2023], the dissociation between language and arithmetic representations [Kisako et al., 2025], and the impact of semantic cues on computation [de Leeuw et al., 2025]. Hanna et al. [2023] identified specific circuits in GPT-2 that compute greater-than comparisons, revealing that MLPs boost probabilities of numerically valid continuations. Kisako et al. [2025] showed that arithmetic equations and general language occupy separate regions in LLM representation space, raising the question of whether French number words activate the "language" region, the "arithmetic" region, or both. Marjieh et al. [2025] used similarity-based prompting to show that LLM number representations blend string-like (Levenshtein distance) and numerical (log-linear distance) structure. Our work provides a new lens on these questions: French vigesimal numbers are simultaneously linguistic (word-form) and arithmetic (encoding $4 \times 20 + 10 + 8$).

Cross-lingual and multilingual representations. Zada et al. [2025] showed that language models trained on different languages converge onto shared conceptual spaces, particularly in middle layers, and that encoding models trained on one language can predict neural activity in listeners of other languages. Xu et al. [2024] evaluated LLMs on numeral conversions across different counting systems, finding that non-standard numeral systems present significant challenges. Our work leverages the French–Belgian French contrast as a within-language controlled comparison: the same numbers, in the same language family, but with vigesimal versus decimal structure. This allows us to isolate the effect of counting system architecture on internal representations while controlling for language-level confounds.

3 Methodology

3.1 Overview

We extract hidden-state representations from MISTRAL-7B v0.1 [Jiang et al., 2023] for numbers 0–999 presented in four surface formats, then train circular digit probes to recover base-10 digit values from these representations. By comparing probe accuracy across formats, layers, and number ranges, we map how the model internally represents French numbers and whether the vigesimal structure degrades these representations.

3.2 Input Formats

For each integer $n \in \{0, 1, \ldots, 999\}$, we construct four input sentences:

1. **DIGITS**: "The number is n." (e.g., "The number is 98.")
2. **ENGLISH**: "The number is *english*(n)." (e.g., "The number is ninety-eight.")
3. **FRANCE-FRENCH**: "Le nombre est *french*(n)." (e.g., "Le nombre est quatre-vingt-dix-huit.")
4. **BELGIAN-FRENCH**: "Le nombre est *belgian*(n)." (e.g., "Le nombre est nonante-huit.")

The France French format uses the standard vigesimal system for 70–99: *soixante-dix* (60+10) for 70, *quatre-vingts* (4×20) for 80, and *quatre-vingt-dix* ($4 \times 20 + 10$) for 90. The Belgian French format replaces these with decimal equivalents: *septante* (70), *huitante* (80), *nonante* (90). Numbers 0–69 are identical across both French variants.

3.3 Representation Extraction

We load MISTRAL-7B in float16 precision on a single NVIDIA RTX A6000 GPU. For each input sentence, we perform a forward pass and extract the hidden-state vector $\boldsymbol{h}_n^{(\ell)} \in \mathbb{R}^{4096}$ at the *last token position* for each layer $\ell \in \{0, 1, \ldots, 32\}$ (embedding layer plus 32 transformer layers). We use the last token because it aggregates information from the full input via causal attention, following the convention of Levy and Geva [2024].

3.4 Circular Digit Probes

We adopt the circular probing framework of Levy and Geva [2024]. Each integer n can be decomposed into three base-10 digits: hundreds (d_h), tens (d_t), and units (d_u), where $n = 100d_h + 10d_t + d_u$. For each digit $d \in \{0, 1, \ldots, 9\}$, we define a circular target:

$$\text{circle}(d) = \left[\cos\left(\frac{2\pi d}{10}\right),\ \sin\left(\frac{2\pi d}{10}\right)\right]. \tag{1}$$

The full target vector for number n is the concatenation $\boldsymbol{y}_n = [\text{circle}(d_h);\ \text{circle}(d_t);\ \text{circle}(d_u)] \in \mathbb{R}^6$.

The probe is a linear map $\boldsymbol{W} \in \mathbb{R}^{6\times 4096}$ with bias $\boldsymbol{b} \in \mathbb{R}^6$, trained to minimize the mean squared error:

$$\mathcal{L} = \frac{1}{|\mathcal{D}|} \sum_{n\in\mathcal{D}} \left\| \boldsymbol{W}\boldsymbol{h}_n^{(\ell)} + \boldsymbol{b} - \boldsymbol{y}_n \right\|^2. \tag{2}$$

We train one probe per layer per format using Adam [Kingma and Ba, 2015] with learning rate 5×10^{-4}, weight decay 10^{-5}, batch size 128, for 500 epochs. At inference, we predict each digit by computing the angle of the probe output and rounding:

$$\hat{d} = \left\lfloor \frac{10}{2\pi} \cdot \text{atan2}(\hat{s}, \hat{c}) \right\rceil \mod 10, \tag{3}$$

where $(\hat{c}, \hat{s})$ are the predicted cosine and sine components and $\lfloor\cdot\rceil$ denotes rounding to the nearest integer.

3.5 Train/Test Split

We use a stratified random split with 80% train (800 numbers) and 20% test (200 numbers), stratified by hundreds digit to ensure equal representation from each block (0–99, 100–199, . . ., 900–999). The test set contains 60 vigesimal numbers (30%) and 140 decimal numbers (70%), reflecting the natural distribution. We use a fixed random seed (42) for reproducibility.

3.6 Evaluation

Overall accuracy. A number is correctly predicted only if all three digits (hundreds, tens, units) match the ground truth simultaneously.

Per-digit accuracy. We report accuracy for each digit position (hundreds, tens, units) separately.

Vigesimal vs. decimal accuracy. We stratify results by whether the number's last two digits fall in the vigesimal range (70–99) or the decimal range (0–69).

Layer-wise analysis. We train probes at each of the 33 layers and report accuracy curves to identify where numeric representations emerge.

Sensitivity analysis. We repeat training with 5 random seeds (42, 123, 456, 789, 1024) at the best layer for each format and report mean $\pm$ standard deviation.

Statistical tests. We use McNemar's test for paired format comparisons on the same test numbers and Fisher's exact test for vigesimal versus decimal accuracy within a format.

4 Results

4.1 Overall Probe Accuracy

Table 1 presents the main results. All four formats encode recoverable base-10 digit information, with word-form formats achieving substantially higher accuracy than digit strings.

FRANCE-FRENCH achieves 98.0% accuracy despite the vigesimal arithmetic structure of numbers 70–99. This is comparable to ENGLISH (98.5%) and only 1 percentage point below BELGIAN-FRENCH (99.0%). The difference between FRANCE-FRENCH and ENGLISH is not statistically significant (McNemar's $\chi^2 = 0.00$, $p = 1.000$), nor is the difference between FRANCE-FRENCH and

Table 1: Overall probe accuracy at each format's best layer. We report accuracy (all three digits correct) with 95% confidence intervals from the single best seed, and mean ± std across 5 random seeds. Word-form representations significantly outperform digit string representations. Best single-seed result in **bold**.

Format	Best Layer	Accuracy (%)	95% CI	5-seed Mean
DIGITS	31	91.0	[87.0, 94.5]	85.5 ± 1.6
ENGLISH	8	98.5	[96.5, 100]	97.0 ± 1.0
FRANCE-FRENCH	6	98.0	[96.0, 99.5]	95.2 ± 1.9
BELGIAN-FRENCH	6	**99.0**	[97.5, 100]	97.7 ± 0.7

Table 2: Probe accuracy for FRANCE-FRENCH stratified by number range. The *soixante-dix* (70–79) range shows the lowest accuracy, while *quatre-vingts* (80–89) and *quatre-vingt-dix* (90–99) achieve perfect accuracy. Best result in **bold**.

Range	n	Accuracy (%)	Structure
All numbers	200	98.0	—
Decimal (0–69)	140	98.6	Standard
Vigesimal (70–99)	60	96.7	Base-20
70–79 (*soixante-dix*)	17	88.2	$60 + 10 + x$
80–89 (*quatre-vingts*)	22	**100**	$4 \times 20 + x$
90–99 (*quatre-vingt-dix*)	21	**100**	$4 \times 20 + 10 + x$

BELGIAN-FRENCH ($\chi^2 = 0.25$, $p = 0.617$). However, FRANCE-FRENCH significantly outperforms DIGITS ($\chi^2 = 9.39$, $p = 0.002$): of the 18 digit-string errors, 16 were correctly predicted by the French probe.

Across 5 random seeds, FRANCE-FRENCH averages 95.2% ± 1.9%, confirming the robustness of the finding. BELGIAN-FRENCH is the most stable format (97.7% ± 0.7%).

4.2 Vigesimal vs. Decimal Accuracy

Table 2 breaks down FRANCE-FRENCH accuracy by number range. The vigesimal range (70–99) shows slightly lower accuracy (96.7%) than the decimal range (98.6%), but this difference is not statistically significant (Fisher's exact test, $p = 0.585$).

The breakdown by vigesimal sub-range is revealing. The *soixante-dix* (70–79) range drops to 88.2% accuracy, while the *quatre-vingts* (80–89) and *quatre-vingt-dix* (90–99) ranges achieve perfect 100% accuracy. This asymmetry is striking: the 80s and 90s use "quatre-vingt" (4×20) as a stem, which the model handles perfectly, but the 70s use "soixante" (60) plus a teens number, which introduces ambiguity.

In contrast, BELGIAN-FRENCH achieves 100% accuracy on all three vigesimal sub-ranges, confirming that the decimal naming system (*septante*, *huitante*, *nonante*) produces uniformly clean representations.

4.3 Layer-wise Analysis

Figure 1 shows probe accuracy as a function of transformer layer. The most striking finding is the divergence in peak layers between word-form and digit-string formats.

All three word-form formats (ENGLISH, FRANCE-FRENCH, BELGIAN-FRENCH) first exceed 90% accuracy at layer 5 and peak between layers 6 and 8. DIGITS, by contrast, does not reach 90% until layer 31 and peaks at layer 31. This means the model extracts numeric meaning from word-form numbers within the first 15–20% of its depth, while digit-string representations require essentially the full network.

Table 3 summarizes the layer dynamics.

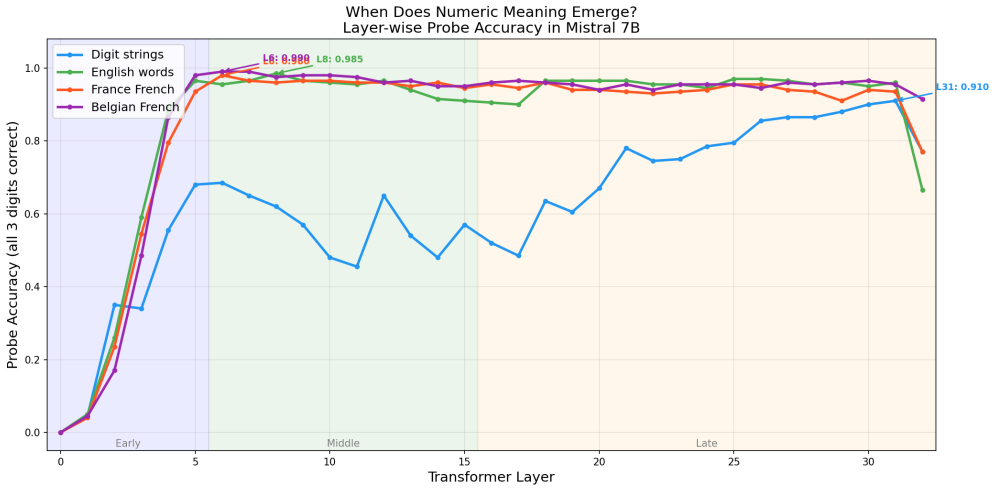

Figure 1: Probe accuracy across transformer layers for all four input formats. Word-form representations (ENGLISH, FRANCE-FRENCH, BELGIAN-FRENCH) peak at early layers (5–8), while DIGITS peaks at the final layer (31). All three word-form formats first exceed 90% accuracy at layer 5.

Table 3: Layer at which probe accuracy first exceeds 90% and the peak accuracy layer for each format. Word-form numbers achieve high accuracy much earlier in the network than digit strings.

Format	**First Layer ≥90%**	**Peak Layer**
DIGITS	31	31
ENGLISH	5	8
FRANCE-FRENCH	5	6
BELGIAN-FRENCH	5	6

4.4 Error Analysis

Table 4 lists all four FRANCE-FRENCH errors. Two errors (73, 76) show a clear pattern: the probe decodes the "soixante" (60) component literally, predicting a tens digit of 6 instead of 7.

The soixante-dix errors are particularly informative. For 73 ("soixante-treize," literally "sixty-thirteen"), the probe predicts 63—recovering "soixante" as 60 and "treize" as 3, but failing to compose $60 + 13 = 73$. The same pattern appears for 76 ("soixante-seize" $\rightarrow$ 66 instead of 76). This suggests that the model's representation partially reflects the compositional arithmetic structure rather than the final composed value.

The "quatre-vingts" (80s) and "quatre-vingt-dix" (90s) ranges show no errors. This may be because "quatre-vingt" functions as a more unitized lexical stem: it always maps to 8 in the tens digit (for 80–89) or 9 (for 90–99 via "quatre-vingt-dix"), whereas "soixante" can map to either 6 (for 60–69) or 7 (for 70–79), creating genuine ambiguity.

By comparison, the 18 DIGITS errors are scattered across the number range without clear arithmetic patterns, likely reflecting tokenization artifacts rather than systematic arithmetic confusion.

5 Discussion

5.1 Interpreting the Results

The model has learned vigesimal arithmetic. Our central finding is that MISTRAL-7B successfully decomposes French vigesimal number words into base-10 digit representations at 98.0% accuracy. When the model processes "quatre-vingt-dix-huit" ($4 \times 20 + 10 + 8$), it arrives at an internal representation from which we can recover the digits $[0, 9, 8]$—the same digits we recover from

Table 4: All FRANCE-FRENCH errors on the test set. The soixante-dix errors reveal partial arithmetic decomposition: the probe recovers "soixante" as 6 in the tens digit rather than the composed value 7.

True	Predicted	French Word	Error Type	Pattern
0	4	zéro	Units: $0 \rightarrow 4$	Edge case
73	63	soixante-treize	Tens: $7 \rightarrow 6$	soixante $= 60$
76	66	soixante-seize	Tens: $7 \rightarrow 6$	soixante $= 60$
116	106	cent-seize	Tens: $1 \rightarrow 0$	Teen confusion

"ninety-eight" or the string "98." This demonstrates that distributional learning on text corpora is sufficient to capture the implicit arithmetic of the French counting system.

Why do word forms outperform digit strings? The fact that all three word-form formats (98.0–99.0%) significantly outperform digit strings (91.0%) is initially surprising. We hypothesize two contributing factors. First, the layer dynamics differ: word-form representations crystallize at early layers (5–6), while digit-string representations only peak at layer 31. The probes trained at the best layer for digit strings may be operating on representations that are still being actively transformed by the final layers, making them harder to decode linearly. Second, digit tokens may be used for diverse purposes beyond pure numeric representation (code, dates, identifiers), creating a more distributed and context-dependent encoding, while number words more consistently signal numeric meaning.

The early-vs-late layer phenomenon. The divergence in peak layers between word-form (layers 5–6) and digit-string (layer 31) representations is our most striking finding. This pattern is consistent across all three word-form conditions, ruling out language-specific explanations. We propose that number words activate the model's semantic processing pathway early in the network—the model "knows" what number a word represents almost immediately—while digit tokens are processed more symbolically, with their numeric meaning emerging only after extensive computation. This aligns with Kisako et al. [2025], who found that arithmetic and language representations occupy separate regions in LLM representation space.

The soixante-dix fingerprint. The error pattern on 73 and 76 (predicted as 63 and 66) provides a direct window into the model's compositional processing. Rather than treating "soixante-treize" as an opaque lexical item mapping to 73, the model's representation partially reflects the arithmetic decomposition: "soixante" $\rightarrow$ 60, "treize" $\rightarrow$ 13. The probe recovers the tens digit as 6 (from "soixante") rather than 7 (from $60 + 13 = 73$). This suggests the representation encodes a superposition of the compositional structure and the final value, with the compositional signal occasionally dominating.

5.2 Limitations

Single model. We evaluate only MISTRAL-7B v0.1. Different models—particularly multilingual models like BLOOM or CroissantLLM, or models with different tokenizers—may show different patterns. The early-vs-late layer finding should be validated across architectures.

Small vigesimal test set. Our test set contains only 17–22 numbers per vigesimal sub-range (70–79, 80–89, 90–99), limiting statistical power for sub-range comparisons. The difference between vigesimal (96.7%) and decimal (98.6%) accuracy is not statistically significant.

Linear probes. Circular probes are linear, which may underestimate the information available in hidden representations if the encoding is non-linear. MLP probes could potentially achieve higher accuracy but at the cost of interpretability.

Last-token extraction. We extract representations at the last token position, which may miss information encoded at earlier positions within compound French words. Token-by-token analysis of how arithmetic composition unfolds across the word "quatre-vingt-dix-huit" could reveal additional structure.

Template sensitivity. We use a single declarative template per language. Different prompts (e.g., questions, arithmetic contexts) may elicit different representations.

Probing limitations. Probing accuracy demonstrates that information is *linearly decodable* from representations, not that the model *uses* this information in its computations [Belinkov, 2022]. The causal interventions of Levy and Geva [2024] provide evidence for functional relevance, but we do not perform interventions in this work.

6 Conclusion

We investigated how MISTRAL-7B internally represents French numbers, with a focus on the vigesimal counting system used for numbers 70–99. Using circular digit probes across four input formats, we found that French number word representations encode base-10 digits at 98.0% accuracy—comparable to English and significantly higher than digit strings. The vigesimal structure leaves localized fingerprints: errors on the *soixante-dix* (70s) range reveal partial encoding of the arithmetic decomposition ("soixante" $\rightarrow$ 60 rather than the composed value 70+). Most strikingly, number word representations peak at early layers (5–6), while digit string representations peak at the final layer (31), suggesting distinct processing pathways for semantic and tokenized numerical inputs.

Our work demonstrates that distributional learning is sufficient to capture the implicit arithmetic of complex counting systems. The French vigesimal system—where 98 is encoded as $4 \times 20 + 10 + 8$ in the surface form—does not prevent the model from building clean base-10 digit representations.

Several directions remain open. Testing additional models (multilingual LLMs, models at different scales) would establish generality. Causal interventions could confirm that these representations are functionally used. Token-by-token analysis of compound vigesimal words could trace how arithmetic composition unfolds across the input. Finally, extending this analysis to other complex counting systems—Danish (*halvfjerds* = "half-fourth-times-twenty" for 70), traditional Japanese, or Basque—would further illuminate the relationship between linguistic structure and internal numeric representations.

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